Chapter 25
Credit Derivatives and the CDS Market

Chapter 24 treated the credit default swap as a source of information: a spread from which a hazard rate could be extracted. This chapter treats it as a contract, and the difference matters more here than anywhere else in the book.

A CDS does not pay on default. It pays on a **credit event**, which is a list of things written into a legal document. Most of the time the two coincide and the distinction is pedantic. When they come apart, someone loses a great deal of money for reasons that have nothing to do with whether the borrower could pay—in March 2023 roughly CHF 16 billion of Credit Suisse Additional Tier 1 (AT1) instruments were written down to zero while shareholders kept about CHF 3 billion, and no credit event occurred at all. Sections 25.10 to 25.12 are about the occasions on which the two have come apart.

The chapter runs from the contract outward. Sections 25.1 to 25.5 are the single-name contract itself: its legs, its credit events, the 2009 standardization that made the risky annuity the whole of valuation, how a position is marked, and how a defaulted contract is settled. Sections 25.6 to 25.8 build portfolios out of it (index CDS and options on it, index tranches, and basket default swaps) and show that the same 125 issuers can be five different instruments depending on one correlation parameter, which is the bridge to Chapter 26. Section 25.9 is the total return swap, the other contract that separates economics from ownership. Sections 25.10 to 25.12 are the cases in which the definition and the economics came apart, and Section 25.13 turns them into a taxonomy of what a CDS actually insures. Sections 25.14 to 25.17 are the neighboring instruments: convertible bonds, credit-linked notes, hybrid structures, and sovereign CDS paid in a foreign currency. Sections 25.18 and 25.19 close with what the post-crisis reforms did to the market and who trades what is left of it.

Section 9.9 pointed here to show that its wrong-way risk example (sovereign protection bought from a bank in that country) was not hypothetical, and Section 25.17 takes it up. Section 10.10 promised convertible bonds here, and Section 25.14 supplies them, because a convertible is the other major instrument in which credit and an option are fused.

25.1 THE CONTRACT

A credit default swap has two legs on a notional N referencing an entity and a class of its obligations.

The premium leg. The buyer of protection pays the premium quarterly until maturity or a credit event, whichever comes first. The market calls the fixed rate at which it is paid the **coupon**, a word borrowed from bonds for a contract that is not one, and this chapter follows the market. The ISDA definitions do not: there the rate is the **Fixed Rate** and the protection buyer is the Fixed Rate Payer, so a reader who opens the legal document will find no coupon in it, and a confirmation that says Fixed Rate 100 basis points and a trader who says 100 coupon are describing the same term. On a credit event the buyer also pays the accrual from the last

payment date, a feature Example 24.1 valued at about a tenth of a percent of the spread for an investment-grade issuer.

The protection leg. On a credit event, the seller pays $\mathcal{N}(1-R)$, where R is determined by the auction of Section 25.5.

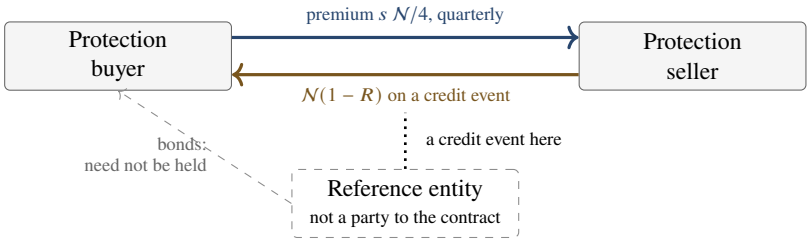


Figure 25.1: The premium and protection legs run between buyer and seller. The dashed links are the ones that may not exist.

Figure 25.1 is the contract, and the two dashed lines are all that is peculiar about it. The reference entity is named by the contract and is party to nothing. It is not consulted, not notified, and receives nothing; it may be unaware that protection on its debt trades at all. A borrower's cost of funds can therefore be quoted in a market it cannot participate in, which is the mechanism behind Section 25.13's question about what a CDS actually insures.

And the link from the reference entity to the buyer is dashed because it need not exist. A protection buyer who holds no bond is not hedging anything, and Section 25.6's index market and Section 25.5's settlement problem both follow from the fact that this link is optional.

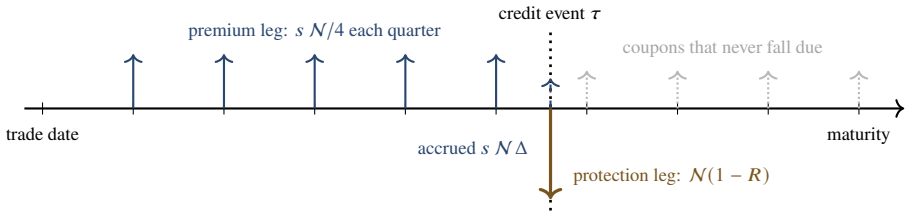


Figure 25.2: Quarterly premium against a single contingent payment. Schematic; the accrual stub is drawn at exaggerated length.

Figure 25.2 puts the same two legs on a calendar. Everything difficult about a CDS is a consequence of the asymmetry in that picture: the premium leg is a stream of small certain payments and the protection leg is one large uncertain one, so the contract's value is dominated by an event whose probability no one observes and whose payout depends on an auction that has not been held.

The buyer of protection is short the credit. They profit if the entity deteriorates, in the same way a put buyer profits if a stock falls, and they need not own the underlying bond—which is the feature that made the instrument useful and the feature that made it controversial.

It completes the market. A bank with a loan it cannot sell can hedge it. An investor who thinks a credit is mispriced can act on that view in either direction. Before CDS, credit was a one-sided market in which the only way to be short was not to own.

It decouples the size of the exposure from the size of the debt. More protection can be outstanding than there are bonds, which matters at settlement (Section 25.5) and which means the instrument's gross notional does not measure what is paid on a credit event, as Section 1.6 warned about derivatives generally.

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25.2 CREDIT EVENTS, AND WHY THE DEFINITION IS THE CONTRACT

The ISDA definitions specify which events trigger payment. For corporates the standard list is bankruptcy, failure to pay, and, depending on the region, restructuring. For financial reference entities the 2014 definitions add governmental intervention, a write-down or bail-in of the reference-level debt imposed by a resolution authority. For sovereigns it adds **repudiation or moratorium** (the government disowns its debt or declares a suspension of payments on it) and, for emerging-market sovereigns, **obligation acceleration**: a default of some other kind on one of its obligations that lets the holders demand repayment at once, before the debt falls due.

Each is defined in detail, and the detail is the contract.

Failure to pay requires a payment actually missed, after any grace period, above a threshold. A payment made late but within the grace period is not a credit event. A payment made by a guarantor is not a credit event.

Bankruptcy is a list of legal states, and different jurisdictions enter them differently. A firm may be economically insolvent for a long time without entering any of them.

Restructuring is the difficult one, and Section 25.10 takes it separately.

Whether a credit event has occurred is decided by a determinations committee. Since 2009 that question has gone to a committee of dealers and investors rather than to a court or the parties to the trade, and its decisions bind the market. This is efficient, because it prevents thousands of bilateral disputes over the same question. It also concentrates the outcome in a body whose members have positions, a governance problem the market long noticed. ISDA's 2025 reform added an independent governance committee and reasoned decisions, but the members of the determinations committees still hold positions.

The gap between economic distress and a contractual credit event is the chapter's recurring theme. A protection buyer who is right about the credit and wrong about the definition receives nothing.

25.3 STANDARDIZATION: THE BIG BANG

Until 2009 CDS traded at par: the coupon was set so the contract had zero value at inception, so every trade had a different coupon and no two positions could be netted.

The 2009 reforms, the "Big Bang," fixed the coupon at a small menu of standard values and moved the difference into an upfront payment:

$$\text{upfront} = \left(s_{\text{par}} - c_{\text{fixed}}\right) \times A \tag{25.1}$$

where A is the **risky annuity**, the present value of one basis point paid until default or maturity.

For North American corporate issuers the menu is 100 basis points for investment grade and 500 for high yield, and those two are by far the most traded; this chapter's examples use them. The menu is wider elsewhere because the upfront has to stay moderate:

- Europe adds 25 for the tightest credits and 300, 750, and 1,000 for wide ones, so that a distressed issuer is not quoted at half its notional upfront;
- sovereign CDS trades mostly on 25 or 100;
- loan CDS uses 100, 250, and 500;
- Japan uses 25 alongside 100 and 500.

A contract's coupon is therefore one of a small number of round numbers chosen to sit near the issuer's spread, and every quoted spread is converted to an upfront against whichever of them the contract carries.

The risky annuity carries the whole valuation, so it is worth building once.

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A CDS has two legs, and both are expectations over the survival curve Q of Section 24.1. Write $t_1 < \cdots < t_n$ for the premium dates, Δ_i for the accrual fraction of the period ending at t_i, and $D(t)$ for the risk-free discount factor. The accrual fraction is a day-count fraction, not a quarter of a year: the standard contract accrues Actual/360 between its quarterly payment dates (March 20, June 20, September 20 and December 20, which the CDS market calls IMM dates although they are not the third-Wednesday futures dates of Chapter 6), so Δ_i runs from about 0.2500 to 0.2556 with the calendar and averages about 0.2536, and Example 25.1 measures what setting it to 0.25 costs. The **premium leg** collects the coupon on every date the issuer survives to, plus the part-period accrued if it defaults in between:

$$A = \underbrace{\sum_{i=1}^{n} \Delta_i \, D(t_i) \, Q(t_i)}_{\text{scheduled}} + \underbrace{\frac{1}{2} \sum_{i=1}^{n} \Delta_i \, D(t_i) \big[Q(t_{i-1}) - Q(t_i) \big]}_{\text{accrued on default}} \tag{25.2}$$

per unit of spread. That is the risky annuity. The $1/2$ is the textbook simplification that default lands, on average, halfway through the period; Hull and White (2000) and O'Kane and Turnbull (2003) both use it. The market's own model does not: the ISDA CDS Standard Model, through which every quoted spread is converted to an upfront, integrates the accrued premium over each period against the hazard rate. At quarterly frequency the two agree to a tenth of a percent of the accrual term (0.0012857 against 0.0012866 per unit of spread for the first quarter of Example 25.1), because, conditional on default inside a period, the expected default time sits only about $\lambda\Delta/12$ of the period before its midpoint, and discounting from that date rather than the period end nearly offsets it. The simplification is kept here for the algebra it makes possible in (25.5). The **protection leg** pays $1 - R$ whenever default arrives:

$$V = (1 - R) \sum_{i=1}^{n} D(t_i) \big[Q(t_{i-1}) - Q(t_i) \big] \tag{25.3}$$

Setting the two equal defines the par spread:

$$s_{\text{par}} = \frac{V}{A} \tag{25.4}$$

A CDS spread is a ratio of two present values, not a yield, and the denominator is what makes it behave unlike one. Everything else in this chapter is a consequence of that denominator: the upfront of (25.1), the mark-to-market of Section 25.4, the index wedge of Section 25.6.

Why the credit triangle is nearly exact. Under a flat hazard rate, a flat discount curve, and a constant accrual fraction Δ the sums in (25.2) and (25.3) are geometric and cancel term by term, leaving

$$s_{\text{par}} = \frac{2(1 - R)}{\Delta} \tanh\!\left(\frac{\lambda \Delta}{2} \right) \tag{25.5}$$

Two things follow, and both answer questions the book has already raised. The discount rate has vanished entirely, which is why, with a flat hazard rate, a change in the risk-free curve alone does not move a par CDS spread: the survival probabilities and discount factors appear identically in both legs and divide out. With a sloping hazard curve the discount factors survive only as weights, and the spread barely moves. And as $\Delta \to 0$ the expression becomes $\lambda(1 - R)$, so Section 24.2's credit triangle is the continuous-premium limit of this contract rather than an approximation to it. At quarterly premiums and the 4.167% hazard of Example 25.1 the gap is two thousandths of a basis point. [TN25] does the integrals behind both legs without the half-period rule, derives the closed forms under flat curves, and sets the three payment conventions (at default, half-period, period end) side by side.

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Example 25.1 takes a single contract from a quoted spread to a cash amount; the table in Example 25.2 is the same calculation repeated across spreads.

Example 25.1 Valuing a CDS from first principles

$10 million of five-year protection on an issuer quoted at a par spread of 250 basis points, traded against the standard 100 basis point coupon. Recovery is 40%, the risk-free rate is 3% continuously compounded, and premiums are quarterly.

A hazard rate comes first. The credit triangle turns the quote into a default intensity,

$$\lambda = \frac{s}{1-R} = \frac{0.0250}{0.60} = 4.167\%$$

and the survival probability follows as $Q(t) = e^{-\lambda t}$:

Year	1	2	3	5
$Q(t)$	0.9592	0.9200	0.8825	0.8119
Cumulative default	4.08%	8.00%	11.75%	**18.81%**

The premium leg. The buyer pays the coupon quarterly, but only while the issuer survives, and pays the accrued fraction of a period if it defaults part-way through one. Per unit of spread,

$$A = \underbrace{\sum_i \Delta_i D(t_i) Q(t_i)}_{\text{scheduled}} + \underbrace{\sum_i \frac{\Delta_i}{2} D(t_i)\left[Q(t_{i-1}) - Q(t_i)\right]}_{\text{accrued on default}} = 4.1647 + 0.0218 = \mathbf{4.1865}$$

which is the risky annuity, and the 250 bp row of Example 25.2.

The protection leg. The seller pays $1 - R$ of notional when default arrives, discounted to today:

$$(1-R)\sum_i D(t_i)\left[Q(t_{i-1}) - Q(t_i)\right] = \mathbf{10.4662}\% \text{ of notional}$$

The two legs should reproduce the quoted spread. Dividing the protection leg by the annuity gives the 250 bp back:

$$s = \frac{0.104662}{4.1865} = 250.00 \text{ bp.}$$

The exactness is a property of the convention rather than a theorem: with the default payment and the accrued coupon both made at the quarter end the discount factors cancel, and the closed form (25.5) gives 249.998 bp, within two thousandths of a basis point. The accrual term is what makes it work: worth only 0.0218 of the 4.1865, half of one percent of the annuity, but leaving it out puts the par spread at 251.31 bp.

Pay the protection when default actually happens, which is what the contract says, and the leg is worth 10.5056% instead of 10.4662%, the par spread is 250.94 bp, and the upfront becomes 6.3191% ($631,908), some $3,941 more. So the credit triangle is exact for a continuously paid premium and about a basis point light for a quarterly contract.

None of what has been assumed is innocent. Default arrives with constant intensity, so the credit curve is flat. The risk-free rate is flat and independent of the default time, so no wrong-way risk appears. Recovery is known at 40% rather than uncertain and correlated with default. And $\Delta_i = t_i - t_{i-1}$ is taken as exactly 0.25: real contracts accrue Actual/360 between the March, June, September, and December payment dates, making Δ_i nearer 0.2536, which lifts the annuity by 1.44% and, with the quote held at 250 bp, raises the upfront by $8,154. The conventions are worth more than the third decimal place of the model.

Now the cash changes hands. Protection is worth more than the 100 bp coupon stream, and the buyer pays the difference at inception:

$$\text{upfront} = 10.4662\% - 100\,\text{bp} \times 4.1865 = 10.4662\% - 4.1865\% = \mathbf{6.2797}\%$$

or **$627,966** on $10 million. Equation (25.1)'s short form, $(250 - 100)\text{ bp} \times 4.1865$, gives 6.2798%. The par spread's 0.002 bp shortfall from 250 bp, times the annuity, is the 0.0001% between them: $9 on $10 million, $627,975 against $627,966.

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Example 25.2 From a spread to an upfront

Five-year contracts on a 100 bp coupon, recovery 40%, rates 3%:

Par spread	Hazard rate	Risky annuity	Upfront
40 bp	0.667%	4.5513	−2.73%
100 bp	1.667%	4.4428	0.00%
150 bp	2.500%	4.3550	+2.18%
250 bp	4.167%	4.1865	+6.28%
500 bp	8.333%	3.8027	+15.21%
800 bp	13.333%	3.4044	+23.83%

The annuity falls from 4.55 to 3.40 as the spread widens, because a contract on a deteriorating credit is expected to terminate early. The risky annuity is the credit market's duration, and it shrinks as the credit worsens, because default ends the payments rather than only discounting them.

The purpose of the convention is fungibility. With a fixed coupon, standard payment dates, and a standard effective date for the protection, two contracts on the same issuer and maturity differ only in the upfront paid at inception, a cash flow already settled, so they net, they can be cleared, and a position can be closed by trading the offsetting contract. The reform was infrastructure, and it did more for the market's resilience than any pricing insight in this chapter.

25.4 MARKING A POSITION

Under the standard contract a protection position's value is the difference between the current par spread and the fixed coupon, multiplied by the current risky annuity,

$$\text{MTM} = \left(s_t - c\right) A_t \tag{25.6}$$

where c is the standard coupon the contract pays (100 or 500 bp for a North American corporate) and not the par spread at which it was traded. That spread has already been settled: the buyer paid or received the upfront of (25.1) on the trade date, and the cash is gone. What remains is a contract paying c, and (25.6) is what it is worth. Two things are easy to miss: that c is the coupon, not the traded spread, and that *both* objects on the right carry the subscript t. Example 25.3 shows what each mistake costs.

Example 25.3 The annuity moves too

A desk bought five-year protection when the par spread was 120 bp, on the standard 100 bp coupon. The five-year risky annuity at 120 bp was 4.4074, so by (25.1) it paid $(120 - 100)\text{ bp} \times 4.4074 = 0.88\%$ of notional upfront, \$88,148 on \$10 million. A year later the four-year par spread is 300 bp.

What the desk holds is a four-year contract paying 100 bp, and the four-year risky annuity *at the new spread* is 3.4103, so

$$\text{MTM} = (300 - 100)\text{bp} \times 3.4103 = 6.82\% \text{ of notional},$$

or \$682,067, what a dealer would pay to take the contract over. The gain since inception is that less the upfront paid, $6.82\% - 0.88\% = 5.94\%$, or \$593,919, before the year of 100 bp coupons the desk has paid to carry the position.

Two errors are common. Marking against the traded spread, $(300 - 120)\text{ bp} \times 3.4103 = 6.14\%$, mixes a cash flow that has already happened into the value of what remains: the 120 bp was settled in the upfront and does not enter the mark. And using the annuity at the old spread (the four-year annuity at 120 bp is 3.6118) gives $(300 - 100)\text{ bp} \times 3.6118 = 7.22\%$, an overstatement of 5.9%. The annuity must be computed on the current curve, because the premium stream actually received will be the one the market now expects.

The direction of the second error is systematic. As a credit deteriorates the annuity shortens, so protection gains less than a naive duration calculation suggests. The position is concave in the

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spread, and the concavity works against the protection buyer. A desk sizing a hedge on duration alone will be under-hedged exactly when the hedge is working.

The standard risk measure is the **CS01**, the change in value for a one-basis-point rise in the spread. It equals the risky annuity times one basis point only when the spread is at the coupon; here the annuity's fall as the spread widens makes the CS01 $3,195 per $10 million, 6% below the $3,410 that the annuity alone suggests, so it must be recomputed at the current spread.

25.5 SETTLEMENT AND THE AUCTION

When a credit event occurs, the contract must be settled, and the mechanism has a history.

Physical settlement was the original design: the protection buyer delivers a defaulted bond and receives par. It works only if there are enough bonds, and Section 25.1 noted that protection outstanding can exceed the debt outstanding. When it does, protection buyers bid for deliverable bonds and their price is driven *up*, which reduces the payout on exactly the contracts that are supposed to be paying out. This happened, and it is why the mechanism changed.

The auction replaced it. Run under separate protocols before 2009 and written into the standard contract by the 2009 reforms, a two-stage auction determines a single recovery rate for the whole market, and all contracts settle in cash against it. The first stage collects dealer markets and physical settlement requests to establish an initial price and a **net open interest** (the amount of bonds still wanted for purchase or offered for sale once offsetting physical requests are matched); the second runs a limit-order auction to clear that interest, and the clearing price is the final recovery.

The design solves the squeeze and creates a different problem. A single number now determines the payout on every contract, so the auction's outcome is worth a great deal to a great many people, and the incentive to influence it is proportionate. The mechanism's defenses are that it is open, that participation is broad, and that the two stages make manipulation expensive—not that it is impossible.

And the auction recovery is not the eventual recovery. It is the market's price for the defaulted claim a few weeks after the event, which is a forecast made under duress. Realized recoveries after a completed restructuring have differed from auction recoveries substantially in both directions, which matters for Chapter 24's Example 24.3: the R used to convert spreads into hazard rates is a convention, and even the observed R at settlement is a market price rather than a realized outcome.

The Lehman Brothers auction of October 2008 is the standard illustration. It set the recovery at 8.625 cents on the dollar, so sellers of protection owed 91.375% of face, yet the Depository Trust and Clearing Corporation (DTCC) reported about $72 billion of gross notional on Lehman and a net transfer of only about $5.2 billion, against rumors near $400 billion. That is netting, as Section 1.6 said, not a failure of the contracts.

25.6 INDEX CDS

Most CDS risk trades through indices (CDX in North America, iTraxx in Europe), each a standardized portfolio of equally weighted issuers (125 for the investment-grade indices, which the examples assume; 100 for CDX high yield and 75 for iTraxx Crossover), rolling to a new series every six months.

The index is built by rule, not by valuation. An administrator selects the constituents at each roll, weights them equally, and fixes the coupon (100 bp for the investment-grade indices, 500 for high yield) for the life of the series. For CDX Investment Grade the constituents are 125 North American issuers rated investment grade and ranked by dealers' liquidity polls; the high-yield and European indices apply the same procedure to a different universe. A new series, which becomes the *on-the-run* series, is launched every March and September; the old one

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continues to trade as an *off-the-run* series. A constituent that suffers a credit event between rolls is not replaced. Its $1/125$ share is settled through the auction of Section 25.5 (the protection buyer receives $(1-R)/125$ of the index notional), and it drops out: the contract's notional is scaled by an **index factor**, $124/125 = 0.992$ after one event, the coupon is paid on the reduced notional from then on, and the series is relabeled a new version so that contracts traded before and after the event remain the same instrument. The survivors carry on unchanged to the series' own maturity; the next roll launches a new series and neither alters nor terminates the old one. High-yield indices, with 100 constituents and more frequent events, see the factor move most; investment-grade series often finish their life at a factor of one.

An index CDS is therefore a single contract paying protection on each constituent as it defaults, with the notional reduced accordingly. It is the credit market's principal liquidity: bid–ask spreads on the index are a fraction of those on constituents, and a desk hedging a portfolio will usually hedge with the index and accept the basis.

Example 25.4 Why the index does not trade at the average spread

Consider a 125-issuer index whose constituents average 80 bp: 115 issuers at 55 bp and 10 distressed issuers at 367.5 bp.

The arithmetic average is 80.0 bp. The spread that makes the index's protection and premium legs balance is **77.3 bp**.

The 2.7 bp difference is arithmetic, not a mispricing. Both legs are additive across constituents. The index's protection leg is the sum of the constituents' protection legs, $\sum_j V_j$, each on its $1/125$ share; and because every constituent pays the same coupon until its own default, the index's premium leg per unit of coupon is the sum of their risky annuities, $\sum_j A_j$. Applying (25.4) to the two sums gives

$$s_{\text{index}} = \frac{\sum_j V_j}{\sum_j A_j} = \frac{\sum_j A_j\, s_j}{\sum_j A_j} \tag{25.7}$$

so the index's fair spread is the annuity-weighted average of its constituents' par spreads, not the arithmetic one. This is what the constituents imply the index should be worth, its *intrinsic* level, rather than how the index is built or quoted: the traded index has its own price, and the standard model converts that quote to an upfront with a single flat hazard curve for the whole index rather than constituent by constituent. A wide-spread issuer has a shorter risky annuity (Example 25.2's effect), so it carries less weight in (25.7) than in a simple mean, and the index trades below it. The wider the dispersion, the larger the gap.

The difference between the level at which the index actually trades and the intrinsic level (25.7) computed from its constituents is the **index skew**. It is a real trading opportunity when it is large enough to cover the cost of trading 125 individual issuers against one index. It widens when the index is used heavily as a macro hedge, because that demand moves the index without moving the constituents. This is Section 19.7's index protection demand appearing in a different market.

Options on a CDS. A **payer option** gives the right to buy protection at a fixed spread K on the expiration date T, and a **receiver option** the right to sell it. With the forward risky annuity A as numeraire (the premium leg of (25.2) starting at T, valued today), a payer on a forward spread s_F with volatility σ is worth $A\,[\,s_F N(d_1) - K N(d_2)\,]$, where $d_{1,2} = [\ln(s_F/K) \pm \sigma^2 T/2]/(\sigma\sqrt{T})$; this is a payer knocked out by default before T, and one that is not adds the **front-end protection**. With $s_F = 250$ bp on five-year protection starting at $T = 1$, so that $A = e^{-(0.03+0.0417)} \times 4.1865 = 3.897$, $K = 300$ bp and an illustrative $\sigma = 60\%$, $d_1 = -0.004$, $d_2 = -0.604$ and the payer is worth $3.897\times(250\times0.4985 - 300\times0.2730) \approx 166.5$ bp of notional. Macro funds buy payers on the investment-grade index to cap the cost of a widening at the premium; dealers carry the volatility.

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25.7 INDEX TRANCHES

Slice the index's loss into layers and sell each separately. A tranche $[a, b]$ absorbs portfolio losses between a and b percent of notional: a is its **attachment point**, the pool loss at which it begins to take losses, b its **detachment point**, the pool loss at which it is exhausted, and the width $b - a$ is its own notional as a fraction of the pool's. The 0–3% **equity** tranche takes the first losses; the 15–30% **senior** tranche only suffers after everything below it is exhausted; the layers in between are the **mezzanine** tranches.

Section 24.13 established the essential point: correlation leaves the portfolio's expected loss unchanged and redistributes it between the layers. The procedure comes before the numbers, because a tranche is valued the same way whatever the pool.

Valuing a tranche. Take a pool of n equally weighted issuers with a common default probability $\pi(t)$ by each date t, recovery R, and the one-factor copula of Section 24.13 with correlation ρ. Four steps take a tranche from that description to a par spread, and only the second of them is where the model lives.

Step 1: the tranche's loss as a function of the pool's. Write $K(t)$ for the number of defaults by t and $L(t) = (1 - R)\,K(t)/n$ for the pool's loss fraction. A tranche $[a, b]$ has lost, as a fraction of its own notional,

$$T_{[a,b]}(L) = \frac{\min\{\max(L - a,\ 0),\ b - a\}}{b - a}, \tag{25.8}$$

zero until the pool's loss reaches the attachment point a, then rising one for one until the detachment point b exhausts it. Its remaining notional at t is $1 - T_{[a,b]}(L(t))$.

Step 2: the pool's loss distribution at each date, conditional on the factor. Given the common factor M (the standard normal variable that stands for the state of the economy in Section 24.13's copula), each issuer has defaulted by t independently with probability

$$p_t(M) = N\!\left(\frac{N^{-1}\big(\pi(t)\big) - \sqrt{\rho}\,M}{\sqrt{1-\rho}}\right), \tag{25.9}$$

which is equation (24.12) with the threshold set at date t's default probability, so $K(t) \mid M$ is binomial with parameters n and $p_t(M)$. The tranche's conditional expected loss is therefore a finite sum,

$$\mathbb{E}\big[T_{[a,b]}(L(t)) \,\big|\, M\big] = \sum_{k=0}^{n} \binom{n}{k} p_t(M)^k \big(1 - p_t(M)\big)^{n-k}\, T_{[a,b]}\!\left(\frac{(1-R)\,k}{n}\right). \tag{25.10}$$

Step 3: integrate out the factor. The unconditional expected tranche loss at t is

$$E_{[a,b]}(t) = \int_{-\infty}^{\infty} \mathbb{E}\big[T_{[a,b]}(L(t)) \,\big|\, M\big]\,\phi(M)\,dM, \tag{25.11}$$

where ϕ is the standard normal density, which is equation (24.13) with the tranche loss as the integrand. In practice the integral is a Gauss–Hermite quadrature (a weighted sum of the integrand at a few dozen fixed values of M) repeated at every payment date.

Step 4: the two legs. On payment dates $t_1 < \cdots < t_m$ with accrual fractions Δ_j and discount factors $D(t_j)$, the protection leg pays the increase in the expected tranche loss over each period, and the premium leg pays the spread on the expected remaining notional, with the half-period accrual of (25.2) on the part lost during the period, which is the same as paying it on the

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expected remaining notional averaged over the period:

$$
\begin{aligned}
V_{[a,b]} &= \sum_{j=1}^{m} D(t_j)\big[E_{[a,b]}(t_j) - E_{[a,b]}(t_{j-1})\big], \\
A_{[a,b]} &= \sum_{j=1}^{m} \Delta_j\, D(t_j)\Big[1 - E_{[a,b]}(t_j) + \frac{1}{2}\big(E_{[a,b]}(t_j) - E_{[a,b]}(t_{j-1})\big)\Big] \\
&= \sum_{j=1}^{m} \Delta_j\, D(t_j)\Big[1 - \frac{1}{2}\big(E_{[a,b]}(t_{j-1}) + E_{[a,b]}(t_j)\big)\Big],
\end{aligned}
\tag{25.12}
$$

and the par spread is $s_{[a,b]} = V_{[a,b]}/A_{[a,b]}$, exactly as in (25.4) with the tranche's expected loss in the place the default probability occupied for a single issuer. Equity tranches are quoted, like the CDS of distressed issuers, as an upfront against a fixed running coupon of 500 basis points through (25.1); that was the equity convention from the first standardized index tranches, and since the 2009 move to fixed coupons the other tranches have been quoted the same way, as an upfront against 100 or 500 basis points.

Everything in Steps 1, 3 and 4 is bookkeeping that any model must do; Step 2 is where the Gaussian copula enters, and replacing it (with random recovery, a heavier-tailed factor, or a different copula) changes only the conditional probability (25.9). The same is true of the pool's description. Equal weights are exact for the standard indices, but a common $\pi(t)$ is a simplification: a dealer's model gives each constituent its own hazard curve, bootstrapped as in Section 24.3, and the binomial in (25.10) becomes a recursion that adds the constituents one at a time (Andersen, Sidenius, and Basu, 2003; Hull and White, 2004), with nothing else in the four steps moving. Example 25.5 runs the four steps on one pool and tabulates what comes out at the five-year date.

Example 25.5 The same portfolio, five different instruments

A 125-issuer index, each issuer with a 7% five-year default probability and 40% recovery. The portfolio's expected loss is $0.07 \times 0.60 = 4.200\%$ of notional at every correlation, because expectation is linear: the pool's loss is the average of 125 indicator variables times $1 - R$, the expectation of each indicator is its own default probability, and no joint probability (hence no ρ) enters the sum.

The tranche numbers come from the copula exactly as Example 24.8 applied it, with a loss given default and a tranche attached. Each default costs $1 - R = 0.6$ of its $1/125$ share, so the pool's loss fraction is $L = 0.6K/125$ for K defaults, and a tranche $[a, b]$ loses $T_{[a,b]}(L)$ of (25.8). This is Step 1. At $\rho = 0.30$ the threshold for a default probability of 7% is $N^{-1}(\pi) = -1.4758$; conditional on the common factor M, defaults are binomial with parameters 125 and $p(M)$, which is Step 2; the conditional pieces (the pool's expected loss and, from (25.10), the expected loss of the equity and senior tranches as fractions of their own notional) are:

M	$p(M)$	$\mathbb{E}[K \mid M]$	$\mathbb{E}[L \mid M]$	$\mathbb{E}[T_{[0,\,0.03]} \mid M]$	$\mathbb{E}[T_{[0.15,\,0.30]} \mid M]$
−3	57.9%	72.4	34.8%	100.0%	99.7%
−2	32.5%	40.6	19.5%	100.0%	30.1%
−1	13.4%	16.7	8.0%	100.0%	0.0%
0	3.9%	4.9	2.3%	71.8%	0.0%
1	0.8%	1.0	0.5%	15.6%	0.0%
2	0.1%	0.1	0.1%	2.1%	0.0%

Integrating each column against $\phi(M)$, which is Step 3 and (25.11), gives the pool 4.200%, the equity tranche 62.59% and the senior tranche 1.88% (the $\rho = 0.30$ row below). The columns show why the two tranches move in opposite directions. The equity tranche is wiped out in every state

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from $M = -1$ down, so it gains nothing further from the bad states that correlation makes more likely, while it is spared in the good states that correlation also makes more likely. The senior tranche loses only in states below about $M = -1.6$, which give it a 4.97% chance of any loss; at $\rho = 0$ the factor moves nothing and that chance is essentially zero. Raising ρ moves probability into those states, and that is the whole of the redistribution.

Expected loss as a percentage of each tranche's own notional, with the 30–100% layer that absorbs everything above the senior tranche:

ρ	0–3%	3–7%	7–10%	10–15%	15–30%	30–100%
0.00	95.76%	32.73%	0.59%	0.00%	0.00%	0.00%
0.10	81.81%	33.49%	9.44%	2.16%	0.10%	0.00%
0.20	71.40%	31.51%	13.50%	5.48%	0.78%	0.00%
0.30	62.59%	29.30%	15.21%	7.87%	1.88%	0.03%
0.50	47.36%	24.75%	15.81%	10.40%	4.27%	0.22%
0.80	26.59%	17.11%	13.35%	10.75%	6.96%	1.05%

Weight each column by its layer's width (0.03, 0.04, 0.03, 0.05, 0.15, 0.70) and every row adds to 4.200%: correlation redistributes the loss and cannot create or destroy any of it. The last layer is what makes the accounting close, carrying nothing at low correlation and, at $\rho = 0.80$, about a sixth of the pool's expected loss.

The equity tranche's expected loss falls from 95.8% to 26.6% as correlation rises; the senior tranche's rises from zero to 6.96%. *Selling equity protection is buying correlation; selling senior protection is selling it.*

The senior tranche is where the danger is, and the right panel of Figure 25.3 draws it on its own scale. A senior tranche valued on a correlation of 0.10 (expected loss 0.098%) and actually facing 0.50 (4.272%) has an expected loss 43 times larger. Nothing about the underlying credits has changed: the same 125 issuers, the same 7% default probabilities, the same 4.2% portfolio loss. Only a parameter that no market quoted has changed.

Step 4, equation (25.12), turns the expected losses into prices. Take the default probability to build up at a flat hazard rate fitted to the five-year figure, $\lambda = -\ln(0.93)/5 = 1.451\%$, so that $\pi(t) = 1 - e^{-\lambda t}$; quarterly payments; and 3% discounting. At $\rho = 0.30$ the five tranches' par spreads are

	0–3%	3–7%	7–10%	10–15%	15–30%
Five-year expected loss	62.59%	29.30%	15.21%	7.87%	1.88%
Risky annuity $A_{[a,b]}$	2.86	4.02	4.35	4.50	4.60
Par spread (bp)	2,062	671	318	158	37

The spread is not the expected loss divided by five. The equity tranche's annuity is only 2.86 because most of its notional is expected to be gone before maturity, so the running spread is correspondingly high. That is why the market quotes that tranche as an upfront instead: against the 500 bp coupon, $(0.2062 - 0.05) \times 2.86 = 44.7\%$ of notional at inception. Repeating the calculation for the senior tranche at $\rho = 0.10$ and 0.50 gives 1.9 and 85.5 bp against 37 at 0.30: the 43-fold move in expected loss is a move of about 45-fold in the spread, because the annuity barely changes for a tranche that is rarely touched.

The rating on such a tranche is a statement about ρ. Section 24.10 said the letter did not disclose the correlation assumption. The table says what that omission was worth: the difference between an instrument that almost never loses and one that loses 4.3% of its notional in expectation.

And the market's own correlation number is not a correlation. Quoting a tranche in terms of the ρ that reproduces its price (**implied correlation**) produces different values for different tranches on the same portfolio, which is logically impossible if ρ were a property of the portfolio. The market moved to **base correlation**: instead of each tranche's own ρ, quote the ρ that prices the **first-loss tranche** $[0, D]$ for each detachment point D (the equity tranche widened to detach at D), and price a mezzanine tranche $[a, b]$ as the difference of the two first-loss tranches $[0, b]$

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and $[0, a]$, each at its own base correlation. A first-loss tranche's value is monotone in ρ, so its implied correlation is unique and the curve can be interpolated between quoted detachment points; a mezzanine tranche's is not: Example 25.5's 3–7% tranche loses 32.73% at $\rho = 0$, 33.49% at 0.10 and 31.51% at 0.20, so one price can correspond to two correlations or to none. The device removes the inconsistency by construction without removing the fact that generated it. This is Section 19.1's implied volatility smile again: a quoting convention rather than a belief.

Chapter 26 is about what happened when it was treated as a belief.

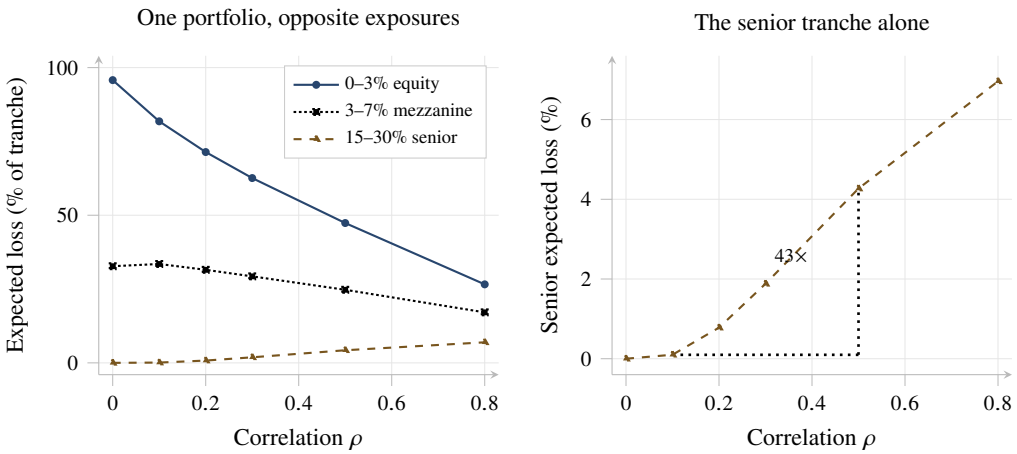


Figure 25.3: Example 25.5's 125-issuer portfolio. The pool's expected loss is 4.200% at every point on both panels.

Figure 25.3 is the picture to carry out of this chapter. The left panel's three curves are one portfolio, so nothing about the credits changes along the horizontal axis, yet the curves go in three different directions; the mezzanine tranche is nearly flat and so says least about ρ. The right panel puts the senior curve on its own scale, where what looked like zero is the forty-three-fold move the example measured.

25.8 BASKET DEFAULT SWAPS

A tranche slices a large portfolio by loss. A **basket default swap** slices a small one (five to ten issuers) by *count*. An nth-to-default basket pays protection on the nth issuer to default and then terminates; the first-to-default basket is the common case, and Section 24.4 has already used it as an instrument whose price depends on the hazard rate at first order.

Nothing new is needed to value one. Write $K(t)$ for the number of issuers that have defaulted by t, whose distribution is Section 24.13's equation (24.13). The basket pays exactly when K crosses n, so it is a credit default swap on a synthetic issuer whose survival probability is

$$Q_n(t) = \mathbb{P}\big(K(t) < n\big) \tag{25.13}$$

Substitute Q_n for Q in (25.2) and (25.3) and the par spread is (25.4) unchanged. The basket is a CDS whose reference entity is the nth order statistic of the portfolio's default times, and once that is seen there is nothing left to derive. Example 25.6 prices all five baskets on one small portfolio.

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Example 25.6 The same five issuers, five different instruments

Five issuers, each trading at 150 basis points with 40% recovery, discounted at 3%, five years, quarterly premiums. Par spreads in basis points:

ρ	1st	2nd	3rd	4th	5th
0.00	750	133	16	1	0
0.15	669	160	34	5	0
0.30	591	179	56	15	2
0.60	439	196	99	47	17
0.90	269	185	138	102	67

At zero correlation the first-to-default costs the sum of its parts. Five issuers at 150 bp give 750 bp, and the basket prices at 750: if defaults never coincide, protection on the first is protection on all five. Correlation is what makes a basket cheaper than the sum, because coincident defaults waste the protection: the second and later defaults arrive when the contract has already terminated.

One entry is worth working in full. Each issuer's hazard rate is $\lambda = 0.0150/0.6 = 2.5\%$ by the credit triangle, so its five-year survival probability is $e^{-5\times0.025} = 0.8825$. At $\rho = 0$ the first-to-default survives only if all five issuers do, $Q_1(5) = 0.8825^5 = 0.5353$. At $\rho = 0.30$, (25.13) is the copula integral $Q_1(5) = \int \big(1 - p_5(M)\big)^5 \phi(M)\, dM = 0.6149$. It is higher because correlation crowds the defaults into the same states and leaves more states with none. Substituting Q_1 for Q at each of the twenty quarterly dates gives a risky annuity of 3.6473 and a protection leg of 0.2157, so $s_1 = 0.2157/3.6473 = 591$ bp, against 3.4665, 0.2600 and 750 bp at $\rho = 0$. The annuity rose and the protection leg fell, and both moved for the same reason.

The opposite movement of the two ends is Section 25.7's, in a different instrument. As correlation rises the first-to-default falls from 750 to 269 bp while the fifth-to-default rises from nothing to 67. A first-to-default is the equity tranche of this portfolio and an nth-to-default with n large is its senior tranche, and the buyer of one and the buyer of the other want opposite things from the same number.

And the five instruments add up. Since $\sum_{n=1}^{5} \mathbf{1}\{K \geq n\} = K$, the five protection legs must sum to the five single-name protection legs at every correlation, and they do, to machine precision. Every default that occurs is paid for exactly once in either portfolio, at the moment it occurs (by the CDS on the issuer that defaulted, or by the basket whose turn it was), so the two sets of cash flows coincide on every path, not merely on average. Correlation redistributes the premium among the baskets and cannot create or destroy any of it, which is Section 24.13's statement about tranches, arrived at from the other direction.

25.9 TOTAL RETURN SWAPS

Section 7.12 introduced the total return swap and said Chapter 27 would examine what it made possible in 2021. The instrument belongs here because it is the principal way credit and equity exposure is taken without ownership.

One party pays the total return on a reference asset (coupons, dividends, and price change) and receives a financing rate. The receiver of the total return has the full economics of owning the asset without owning it, and without paying for it: the position is funded by the swap.

Three properties follow, and they are the same three that recur whenever an instrument separates economics from ownership.

Leverage is set by the margin, not by the asset. A cash buyer of a bond posts its full price. A total return receiver posts initial margin, which may be a small fraction. The economics are identical and the balance sheet is not.

The exposure is not visible in ownership disclosures. The swap counterparty holds the asset as a hedge and appears as the owner. Positions built this way do not show up where people look for them, which is Section 7.12's phrase about leverage accumulating quietly, and Chapter 27's

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case study is what that enabled.

And the risk is counterparty risk presented as market risk. The dealer paying total return is hedged in the market and exposed to the client; if the asset falls, the client owes variation margin, and if the client cannot pay, the dealer holds the asset. The hedge is perfect against the market and worthless against the client, the configuration Chapters 8 and 9 dwelt on.

25.10 RESTRUCTURING

Restructuring is the most contested credit event, and the one where the definition works hardest.

A restructuring (a maturity extension, a coupon reduction, a change in seniority, imposed on holders) is economically a loss but not a missed payment. Whether it should count as a credit event has been answered differently in different places and different decades.

The problem it creates is the cheapest-to-deliver problem in its sharpest form. After a bankruptcy, all senior unsecured obligations are worth roughly the same, so which one is delivered hardly matters. After a restructuring, the entity is still alive, and its obligations trade at very different prices depending on maturity: a thirty-year bond of a distressed but living issuer is worth far less than a one-year bond of the same issuer. A protection buyer could deliver the cheapest, collecting a payout far exceeding their actual loss.

The market's responses were successive restrictions on what could be delivered (limits on maturity, then buckets) and, in North America, largely dropping restructuring as a trigger for corporate issuers. Europe kept it, mainly because bank regulators require it: a hedge that does not pay on restructuring does not obviously transfer the risk of the loan it hedges, and Example 25.7 prices what the clause is worth.

Example 25.7 What the clause is worth

Take an issuer with a 3% annual intensity of bankruptcy or failure to pay, recovering 40%. Suppose including restructuring adds a further 15% to that intensity (0.45 percentage points) but restructuring events recover 70%.

No-restructuring contract	180.0 bp
With restructuring	193.5 bp
Value of the clause	13.5 bp (7% of the spread)

The point is not the number, which depends entirely on the assumed intensities, but that it is small and stable right up until it is neither. When a restructuring is actually negotiated, one contract pays and the other does not, and the gap between them goes from 13 bp to the whole protection payment. A risk system that treats them as substitutes because their historical basis is small has mismeasured a binary exposure as a small one.

25.11 WHEN THE DEFINITION AND THE ECONOMICS COME APART

Three episodes each established something about what a CDS actually insures.

Greece, 2012. The sovereign's debt was restructured through a package that was initially presented as voluntary, and a voluntary exchange is not a credit event. Collective action clauses were then applied retroactively to bind holdouts, and that application—compulsion—triggered the contracts. Protection paid, but only after a period in which it was genuinely unclear whether the largest sovereign restructuring in history would trigger the instrument designed to insure against exactly that.

Manufactured events. In the 2010s several transactions were arranged specifically to trigger or avoid triggering CDS: a distressed borrower persuaded to miss a small payment so that a protection holder would be paid, or refinanced in a way designed to leave protection worthless. ISDA amended the definitions in 2019 to address the most obvious form. The episodes are

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instructive because they show that when a contract references an event rather than a loss, the event can be purchased.

Russia, 2022. Payments on sovereign debt were made in rubles, or blocked by sanctions, or made but unable to reach holders. The question of whether a payment prevented by a third party is a failure to pay had no clean answer, and the determinations process had to resolve a case its drafters had not anticipated. A contract that insures against a borrower's unwillingness or inability is silent about a borrower who is willing, able, and prevented. The committee found a failure to pay in June 2022, on $1.9 million of accrued interest, and sanctions then delayed an auction that set the recovery near 56 cents.

The same gap runs through the corporate market. A liability-management exercise of the kind Section 26.15 describes can cut a lender's recovery while no payment is missed and, in the North American contract, no restructuring counts, so the protection pays nothing, or pays on an obligation whose price has fallen less than the loan's.

25.12 CREDIT SUISSE, MARCH 2023

The most instructive recent case triggered nothing at all, which is the point.

Case Study 25.1 The AT1 write-down and what subordination means

Additional Tier 1 instruments were created after the 2008 crisis to do something specific: absorb losses while a bank is still operating, so that it can be recapitalized without a resolution. They pay a high coupon, and they contain contractual triggers under which they are written down or converted to equity.

In March 2023, as part of the arrangements accompanying Credit Suisse's acquisition, roughly CHF 16 billion of AT1 instruments were written down to zero. Shareholders received approximately CHF 3 billion.

The capital structure was inverted. An instrument senior to equity was extinguished while equity retained value. That is the opposite of the ordering that the word "subordination" is supposed to guarantee.

Three observations follow, and the third is the one for this chapter.

It was contested in law, not only in markets. The Swiss AT1 terms allowed write-down on a viability event determined by the regulator, and holders had been told so. In October 2025, however, the Federal Administrative Court found that no such event had occurred and that the Swiss regulator's order lacked a legal basis; the appeal is pending, and the bonds remain written down. Whichever way it ends, almost no one had priced the possibility, because the instruments had been sold and analyzed as if the ordinary hierarchy applied.

The repricing was immediate and general. AT1 instruments across European banks fell sharply, and several regulators issued statements affirming that in *their* jurisdictions equity would absorb losses first. The market had been pricing a legal ordering that turned out to be jurisdiction-specific.

And no credit event occurred. Credit Suisse did not default; it was acquired. Its senior obligations performed throughout, and when the determinations committee was asked in May 2023 whether the write-down was a governmental intervention, it ruled that it was not: the AT1s ranked below every obligation the contracts reference, subordinated CDS included. So CDS on Credit Suisse did not pay, while holders of an instrument ranking above equity lost everything.

That last sentence is Section 25.2's thesis in its cleanest form. A CDS insures against a list of events, not against loss. A protection buyer who had correctly identified Credit

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Suisse as the weakest large European bank, and had bought protection to express it, would have been right about the institution and received nothing—while an AT1 holder with no view at all lost the entire position.

The general lesson extends past this instrument: wherever a payoff is defined by a legal event rather than by an economic outcome, the document is the instrument, and the residual risk is the risk that the document does not say what everyone assumed.

25.13 WHAT A CDS ACTUALLY INSURES

The episodes of Sections 25.11 and 25.12 are usually told as anecdotes. They are more useful as a taxonomy, because a hedger can check each item in advance.

A protection buyer who wants to be made whole on a credit loss faces four distinct residual risks, and only the last is normally measured.

Definitional risk. The loss occurs and no listed event does. This is the Credit Suisse case, and it is the largest of the four because it is binary and uncorrelated with anything a risk system tracks. The check is to read the credit event definitions against the specific way this borrower could plausibly impose a loss; for a bank, resolution, recapitalization, and default are three routes with three different answers.

Determination risk. An event occurs and the committee rules that it did not, or takes long enough that the position must be marked in the meantime. Greece and the 2022 sovereign cases are both instances. The check is whether the borrower's plausible distress route is one the committee has ruled on before. Novelty is the risk.

Settlement risk. The event is determined, the contract pays, and the auction recovery differs from the loss actually suffered on the hedged asset. A hedger holding a long-dated bond who receives an auction recovery struck on the cheapest deliverable has been made whole on a different instrument.

Basis risk. The protection references an entity and the exposure is to a different one: a subsidiary, a guaranteed affiliate, a proxy issuer entirely, as in the CVA hedging of Section 9.10. This is the one desks do measure, and it is the smallest.

The ordering is the reverse of the attention these risks receive: basis risk is quantified, reported, and hedged, while definitional risk, the one the Credit Suisse case exposed, is not quantified at all.

25.14 CONVERTIBLE BONDS

Section 10.10 deferred convertibles here because they are the other instrument that fuses credit with an option, and the fusion is what makes them interesting.

A convertible is a bond that the holder may exchange for a fixed number of shares, the **conversion ratio**. So it is a corporate bond plus a call on the issuer's own equity, and Section 17.4's dilution analysis applies to that call.

Example 25.8 Decomposing a convertible

Face 1000, 2% annual coupon, five years, conversion ratio 20. The stock is at 40, its volatility 35%, rates 4%, and the issuer's credit spread 300 bp.

How the split is made. The floor is the 2% bond discounted at 4% + 3% = 7%, continuously compounded; the option is a European call on 20 shares struck at the conversion price, valued by the Black–Scholes–Merton formula at the 4% risk-free rate with no dividends ($d_1 = 0.362$, $d_2 = -0.421$, 11.86 a share); and the convertible is their sum. The additive split is an approximation: it treats conversion as exercisable only at maturity, ignores any issuer call, and lets the option survive a default that would extinguish it. Example 25.9 measures each omission. Tsiveriotis and

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Fernandes (1998) is the widely used practitioner model, pricing the debt and equity components on one tree at different discount rates.

Bond floor (straight bond discounted at 7%)	786.14
Conversion value, or *parity* (20×40)	800.00
Conversion price (1000/20)	50.00
Conversion premium over the share price	25.0%
Embedded call on 20 shares struck at 50	237.13
Convertible value	**1023.28**
Equity delta	12.82 shares (64% of the 20)

The instrument's behavior across share prices shows what the issuer sold:

Share price	Convertible	Parity	Delta (shares)
20	827.87	400.00	6.00
30	908.55	600.00	9.95
40	1023.28	800.00	12.82
50	1162.13	1000.00	14.82
70	1484.96	1400.00	17.18

The convertible is bond-like when the stock is low and equity-like when it is high, with the transition governed by the embedded call's delta. That is its selling point to an issuer: a 2% coupon on debt that would otherwise cost 7%, for equity upside above 50: Section 17.9's bond-plus-warrant, with the components fused rather than separable.

Section 13.7 said why the additive split is not a valuation: conversion and call are node-by-node decisions that interact, so a bond and its embedded options cannot be valued separately and added; and credit belongs on the same tree, since a default ends the conversion right too. Hull's construction puts all three on one tree: each step of length Δt the stock moves up by u, down by $d = 1/u$, or the issuer defaults with probability $1 - e^{-\lambda \Delta t}$, leaving the stock worthless and the bond worth its recovery RF:

$$u = e^{\sqrt{(\sigma^2 - \lambda)\,\Delta t}}, \qquad p_u = \frac{e^{r\Delta t} - d\, e^{-\lambda \Delta t}}{u - d}, \qquad p_d = \frac{u\, e^{-\lambda \Delta t} - e^{r\Delta t}}{u - d}, \tag{25.14}$$

chosen so that the stock's expected growth, default paths included, is the risk-free rate and its total variance per unit time, jump included, is σ^2. The credit triangle gives $\lambda = s/(1 - R)$. With conversion ratio n and call price K_c, each node's rollback is

$$V = \max\left\{ nS,\ \ \min\left(e^{-r\Delta t}\left[p_u V_u + p_d V_d + (1 - e^{-\lambda \Delta t})\, RF \right],\ K_c \right) \right\} + \text{coupon due}, \tag{25.15}$$

read from the inside out: the bracket is the continuation value; the issuer calls when continuation exceeds K_c; and the holder converts whenever n shares are worth more than what is left. Without a call the min drops out; with conversion only at maturity the outer max acts only there.

Example 25.9 Pricing the convertible on a tree with default

Example 25.8's bond on a tree of five annual steps, so that the coupon dates are the nodes. From $s = 300$ bp and $R = 40\%$, $\lambda = 0.03/0.6 = 5\%$; then (25.14) gives $u = 1.3090$, $d = 0.7639$, $p_u = 0.5763$, $p_d = 0.3749$, default probability 0.0488 and discount factor $e^{-0.04} = 0.9608$ per step. Figure 25.4 is the whole tree for the callable case.

One node is worth working in full, through the rollback (25.15). At $t = 4$ with the stock back at 40, the surviving successors are $S = 52.36$, where the bond converts at maturity for 1047.19 plus the final coupon, 1067.19, and $S = 30.56$, where it is redeemed for 1020.00 with the coupon; default pays 400. Continuation is $0.9608 \times (0.5763 \times 1067.19 + 0.3749 \times 1020.00 + 0.0488 \times 400) = 977.09$, against $20 \times 40 = 800$ for converting now, so the holder holds, and the node is worth 977.09 plus

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the year-4 coupon, 997.09. The issuer's call at 1100 from year 2 shows up in two forms. At $t = 2$ with $S = 68.54$ continuation is 1426.81, so the issuer calls, and the holder, offered 1100, converts instead for 1370.77: the call forced conversion three years early and cost the holder 56. At $t = 3$ with $S = 52.36$ continuation is 1148.00, the issuer calls, and this time 20 shares are worth only 1047.19, so the holder takes the 1100 in cash and loses 48 against holding on.

	Five steps	**500 steps**
Bond floor (no conversion right)	793.50	795.10
Conversion at maturity only, or at any time	1046.38	1057.82
Conversion at any time, callable at 1100 from year 2	994.47	988.40
Additive split of Example 25.8, for comparison	1023.28	

Three readings deserve comment. The floor is 795 rather than 786 because recovery is modeled as a payment of 400 at default rather than folded into a 7% discount rate. Conversion at maturity only gives 1058 against the additive 1023: 263 rather than 237 for the option, the rest being the higher floor. The default branch makes the call dearer, not cheaper, because a stock that can drop to zero must grow at $r + \lambda$ conditional on surviving (Merton's 1976 jump-to-ruin result), which outweighs the lower diffusive volatility of 26.9%. The American right is worth nothing here, since without dividends the holder never converts early (Section 11.8). What changes the value is the issuer's call, −52 on five steps and −69 converged, because it forces conversion whenever continuing would be worth more than 1100; that is the interaction Section 13.7 said the additive method discards, and it is the largest term.

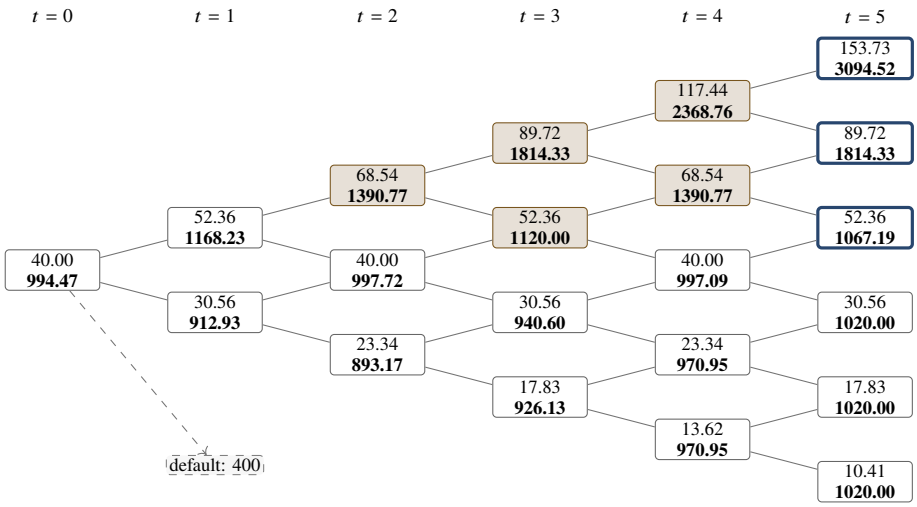


Figure 25.4: Example 25.9's callable convertible on the five-step tree: share price above, bond value (with the coupon paid that date) below. Shaded nodes are called, heavy-bordered nodes converted; the default branch worth 400 exists at every node and is drawn once.

Convertible arbitrage. The classic hedge fund trade buys the convertible and shorts the delta, leaving a position that is:

Long gamma and vega, because the embedded call is now unhedged in second order. This is Chapter 18's dealer position with the sign reversed.

Short credit, because the bond floor is a risky bond.

And long carry, from the coupon less the cost of the short.

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The trade's characteristic failure is that its two exposures go wrong together. A 100 bp widening in the issuer's spread, with the stock unchanged, costs 36.66 on the bond floor (3.58% of the convertible's value). And credit widens precisely when equity volatility spikes, so the position's volatility gain arrives at the same moment as its credit loss.

That would net out if both were realized on the same timetable. They are not. The credit loss is a mark-to-market that arrives immediately; the volatility gain is earned only by gamma-hedging through the move, which requires the fund to still be there. In 2008 many were not, and convertible arbitrage, a strategy that is genuinely hedged in the sense that its Greeks offset, produced one of the worst drawdowns of any hedge fund style. Being hedged in the model and being able to hold the position are different things, which is Section 5.15's limits-of-arbitrage argument in yet another market.

25.15 FUNDED AND UNFUNDED FORMATS

A CDS is **unfunded**: no principal changes hands at inception, and the protection seller's obligation is backed only by their creditworthiness and their Chapter 8 CSA's collateral.

A **credit-linked note** converts the same exposure into a funded one. The investor buys a note, paying cash upfront; the issuer invests the proceeds and pays a coupon; if a credit event occurs on the reference entity, principal is reduced accordingly. Economically the investor has sold protection and prepaid the collateral in full.

The difference is entirely about counterparty risk, and it runs both ways.

The protection buyer is fully secured. They hold the investor's cash, so there is no exposure to the seller at all. That is why credit-linked notes are the format of choice when the seller is an entity no one wants unsecured exposure to.

The investor acquires exposure to the issuer. Their principal now sits with the note's issuer and is invested in something. A credit-linked note is therefore two credit exposures, not one, and the second is easy to overlook because it is not what the note is named after. Several structures in 2008 delivered losses through the collateral rather than the reference entity.

The same funded-unfunded distinction organizes Chapter 26. A cash securitization is funded: investors buy notes and the proceeds buy the assets. A synthetic one is unfunded: investors sell protection on a reference portfolio that stays where it is. The synthetic version can be created without anyone originating a loan, which is why the outstanding amount of synthetic exposure was not limited by the amount of debt.

25.16 HYBRID CREDIT STRUCTURES

The previous section ended by observing that a credit-linked note is two credit exposures rather than one. The same construction generalizes, and it has become the part of the credit market where the modeling is hardest.

A **hybrid** structure makes a payment contingent on a credit event *and* on the level of something else: an interest rate, an exchange rate, an equity index. A note might pay an equity-linked coupon for as long as a reference issuer survives, redeem at a rate-linked amount unless a portfolio takes losses, or pay in a currency the reference entity does not borrow in. Chapter 35 treats the note wrapper and the distribution question; what belongs here is what the credit leg does to the valuation.

The temptation is to price the two legs separately and add, and it does not work. Write τ for the default time and X for the other factor. The payoff is $\mathbf{1}\{\tau > T\}\, f(X)$, and its value involves

$$\mathbb{E}[\mathbf{1}\{\tau > T\}\, f(X)] \;\neq\; \mathbb{P}(\tau > T)\, \mathbb{E}[f(X)]$$

unless default and the factor are independent. The two legs are individually observable (survival probabilities come from the CDS curve of Section 25.3, the distribution of X from its own

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option market), and the quantity that joins them is not observable from either. Example 25.10 puts a number on what that costs.

Example 25.10 The correlation a hybrid needs and no market quotes

A five-year note paying a single 8% coupon at maturity, and paying it only if the reference issuer has survived *and* an equity index finishes above its initial level. Risk-free rate 4%, dividend yield 2%, index volatility 30%, and a CDS curve implying a five-year survival probability of 0.90.

The index condition is met with probability $N(d_2)$, where

$$d_2 = \frac{(r - q - \sigma^2/2)T}{\sigma\sqrt{T}} = \frac{(0.04 - 0.02 - 0.045) \times 5}{0.30\sqrt{5}} = -0.1863$$

so $N(d_2) = 0.4261$.

Couple the two with a Gaussian copula, Section 24.13's device with two variables rather than a pool: default if the credit factor falls below the 10% point of the standard normal, -1.2816, and the index above strike if its factor exceeds $-d_2 = 0.1863$. The coupon is paid when both conditions hold, which is a bivariate normal probability.

ρ	P(coupon paid)	Expected coupon	Effect on the coupon
−0.4	0.3551	2.8408%	−22.7 bp
0.0	**0.3835**	**3.0678%**	**reference**
0.4	0.4091	3.2727%	+20.5 bp

At $\rho = 0$ the probability is exactly 0.90×0.4261, which is the control on the calculation. At a correlation of 0.4 the expected coupon moves by about **20 basis points**—on a trade whose two legs are each quoted to a fraction of a basis point.

The realistic sign is positive. Equity levels and credit quality move together, so the coupon tends to pay in the states where the issuer also survives, and the note is worth *more* than the independent calculation says. A desk that prices the credit leg off the CDS curve and the equity leg off the volatility surface and multiplies the two is therefore not making a small error in an unknown direction; it is understating the note.

Three things follow.

The correlation cannot be implied. Vanilla equity options imply the index's volatility and CDS implies the issuer's hazard rate, and no liquidly traded instrument implies the dependence between them. It is estimated from history, which is Section 26.6's position on default correlation, arrived at in a market that was supposed to have learned that lesson.

The bounds are wide but not useless. Any joint distribution with these marginals pays the coupon with probability between $\max(0.90 + 0.4261 - 1, 0) = 0.3261$ and $\min(0.90, 0.4261) = 0.4261$, so the note's value is confined to a band of $(0.4261 - 0.3261) \times 8\%$, or 80 basis points of coupon, whatever the dependence. That width is $1 - 0.90$ times the coupon and nothing else: for any index probability between 0.10 and 0.90 the two bounds differ by exactly the default probability, so the band is set by the credit leg alone. A reserve sized to the width of that band is defensible without agreeing on a number, which is the same response Section 34.7 offers to model risk in exotics.

And the hedge is cross-gamma. The sensitivity that matters is not delta to the index or spread sensitivity to the issuer but the cross term between them, and there is no instrument whose payoff is that cross term. It is hedged by rebalancing two positions, which means the hedging error is a function of realized rather than implied dependence.

Who trades them, and why. The buyer is an investor who wants a yield the single-factor version cannot produce: attaching a second condition to a coupon raises it, as the range accrual of Section 33.6 does with a rate, because the investor has sold something. The seller is one of the few dealers that can warehouse and partially hedge the cross-risk. Insurers and pension funds buy where the structure lets them reach a yield target within a ratings constraint, the same demand that Chapter 26's senior tranches met.

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What has changed. Section 9.9's wrong-way risk is the same mathematics as this section's correlation: an exposure that grows as the counterparty's credit deteriorates. What has changed is that the difficulty has migrated from the valuation adjustment into the products themselves. A hybrid note needs a model in which credit, rates and an equity index move together, calibrated to three separately quoted markets, and the number it produces is the price the client pays.

25.17 SOVEREIGN CDS AND THE CURRENCY OF THE PAYOUT

Section 24.12 established that sovereign credit is different in kind. The CDS market adds a feature that has no corporate analog and that connects directly to Chapter 16.

Sovereign CDS is usually denominated in a currency not the sovereign's own. Protection on an emerging market sovereign trades in dollars; protection on a euro-area sovereign has traded in dollars and euros. The reason is obvious: protection paying in a currency the defaulting sovereign controls is worth less than it appears. The gap is priced.

The two contracts differ by what Chapter 16 would call a quanto effect. A euro-denominated contract on a euro-area sovereign pays euros in a state of the world in which the euro may have depreciated sharply, precisely because that sovereign defaulted. The dollar contract pays the same notional in a currency that has not moved, so it is worth more, and the **quanto spread** between the two is the market's price for devaluation conditional on default.

It is a jump-at-default expectation. If the dollar contract trades meaningfully above the euro one, the market is pricing a substantial currency move conditional on the credit event, not a general currency view, since an unconditional move affects both contracts alike.

During the euro-area crisis it was a market-implied redenomination probability. The quanto spread on peripheral sovereigns widened when the possibility of leaving the currency union was being discussed and narrowed when it receded. A market with no instrument for pricing the breakup of a currency union priced it anyway, in the basis between two CDS contracts.

The pricing requires exactly the machinery of Section 16.7: a correlation between the credit event and the exchange rate, entering as a quanto adjustment. It is this book's plainest case of a credit instrument and a currency instrument that must be priced jointly (the credit-and-currency hybrids of Section 25.16 are the bespoke version).

The seller matters as much as the currency. Section 9.9's textbook case of wrong-way risk is protection on a sovereign bought from a bank in that country: the contract pays in exactly the states in which a bank holding its own sovereign's bonds, and funded in its currency, is least able to pay. In the euro-area crisis of 2010–2012 the case was live rather than hypothetical: Italian and Spanish banks held large amounts of their own governments' bonds, and their CDS spreads rose and fell with their sovereigns', so protection on Italy sold by an Italian bank was weakest exactly when it would have paid. Protection on a sovereign is therefore worth most when it is bought from a seller outside the jurisdiction and fully collateralized, and a desk that buys it from a domestic bank has to price that correlation into its CVA.

25.18 THE MARKET AFTER THE REFORMS

The instrument this chapter has described trades in a market a small fraction of the size of the one that made it famous.

Case Study 25.2 The market that shrank

Gross notional outstanding peaked at about $58 trillion at the end of 2007 and fell by roughly an order of magnitude over the following decade; at the end of 2025 it stood at $11 trillion, 70% of it cleared. Three forces did it, and they are worth separating because only one of them reduced risk.

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Compression removed offsetting trades. A dealer with a bought and a sold contract on the same issuer, maturity, and coupon holds two contracts and no risk. The Big Bang's fixed coupons and standard dates made those contracts fungible, and multilateral compression services then tore them up in bulk. This eliminated enormous gross notional and no net exposure at all.

Clearing concentrated and netted the rest. Index CDS moved to central clearing under the mandates of Chapter 2, which nets multilaterally rather than bilaterally and shrinks the reported figure further.

And single-name activity genuinely declined. This is the force that reduced risk, because fewer net positions are now taken, and it reduced the market's usefulness with it. Capital charges on dealer inventory rose, the instrument acquired a reputational cost after the crisis, and several large participants withdrew. Liquid single-name CDS now exists for a few hundred reference entities, which Section 9.10 already flagged as the reason most CVA is hedged by proxy.

Against: a bank hedging a loan to a mid-sized borrower usually cannot buy protection on it, so the risk-transfer function that justified the instrument works only for the largest issuers. The market that remains serves index macro hedging and the CVA desks of Chapter 9 better than it serves the original purpose.

For: the market that remains is standardized, cleared, compressed, and settled by auction, and none of those was true in 2007. The instrument that failed in the crisis and the instrument that exists now differ in every respect that mattered then.

The reforms traded breadth for robustness. That is probably the right trade, and it is a trade rather than a free improvement.

25.19 WHO TRADES CREDIT DERIVATIVES, AND WHY

Banks buy protection to hedge loan books and to reduce their capital, and this was the instrument's original purpose. Under the substitution rule of Section 27.11 the hedged loan takes the protection seller's risk weight, so the capital it saves can justify a hedge on its own (Example 27.6).

XVA desks buy protection on their counterparties, as Section 9.10 described. This is the largest structural demand for single-name protection, and it is why the market's liquidity is concentrated in the few hundred issuers that are also large derivatives counterparties.

Asset managers use index CDS to adjust credit exposure faster and more cheaply than by trading bonds. Selling index protection is a synthetic long credit position that requires no cash and no bond inventory.

Macro funds buy index protection as a hedge against a general deterioration, and this demand is what moves the index skew of Example 25.4.

Basis traders arbitrage the CDS–bond relationship of Section 24.15, subject to the funding constraint that section identified.

Correlation desks and hedge funds trade the equity and mezzanine tranches against the index or single names, as in the 2005 trade of Section 26.7; insurers, pension funds, and banks held the senior layers for a yield above their rating, the demand Chapter 26's senior tranches met. **Basket default swaps** let an investor earn more than any one issuer's spread for bearing the first default among a handful of names; the dealer buying the protection hedges with the constituents and keeps the correlation exposure. **Hedge funds** receive total return to hold financed positions they would otherwise fund and disclose (Section 7.12), and buy most new convertibles, alongside long-only convertible funds, for the arbitrage of Section 25.14; issuers are growth companies that would pay far more for straight debt.

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The market's structural position is that dealers are net sellers of protection to banks and buyers from asset managers, with the imbalance hedged in the index. A general widening therefore produces a demand for the same hedge from everyone at once. That is Chapter 22's crowding argument transplanted to credit, and Chapter 26 records the last time.

SUMMARY

A CDS pays on a credit event, not on default. When the two differ, the contract's definitions govern, and a protection buyer who is right about the credit and wrong about the definition receives nothing. Since 2009 a determinations committee of market participants, not a court, rules on whether an event occurred.

The 2009 standardization fixed the coupon at a few standard values (100 or 500 basis points for North American corporates) and moved the difference between the coupon and the issuer's spread into an upfront payment. The risky annuity falls as the spread widens, because a deteriorating credit's premium stream is expected to be cut short, so a hedge sized on duration is under-hedged exactly when it is working.

Physical settlement failed because protection outstanding can exceed deliverable debt, and the auction replaced it with a single market-wide recovery rate, a forecast made under duress, not the realized recovery. Slicing an index into tranches redistributes a fixed expected loss: as correlation rises, loss moves out of the equity tranche and into the senior one, so a senior tranche valued at a low correlation and facing a high one carries many times the expected loss it was priced for. Base correlation removes the inconsistency between the tranches' implied correlations without removing its cause. An index trades at the annuity-weighted, not the arithmetic, average of its constituents' spreads, and correlation lowers the price of a first-to-default basket while raising that of the last.

Restructuring is the contested credit event because it leaves the issuer alive with obligations at very different prices. Greece in 2012, the manufactured events of the 2010s, Russia in 2022, and Credit Suisse in March 2023, whose AT1 bonds were written to zero with no credit event, all show that where a payoff is defined by a legal event, the document is the instrument. The market shrank by an order of magnitude after the crisis, trading breadth for robustness.

A credit-linked note is a funded CDS with a second exposure to the note's issuer; a total return swap gives the economics of an asset without ownership; sovereign CDS pays in a currency the sovereign does not control, and the quanto spread is the price of devaluation conditional on default. A convertible is a corporate bond plus a call on the issuer's equity, but default, the conversion right, and the issuer's call interact and must be priced together on one tree. Convertible arbitrage is long gamma and vega, short credit, and long carry, with the credit loss marked immediately and the volatility gain earned only by hedging through the move. A hybrid coupon that needs both survival and an equity level cannot be priced as a product of marginals; the unquoted dependence moves it within a band set by the credit leg alone.

FURTHER READING

The ISDA credit derivatives definitions are the primary source, and Iwashita's OpenGamma note on single-name CDS conventions collects the coupon menus, day counts, and roll dates that the definitions leave to market practice; anyone who intends to rely on a CDS should read the credit event definitions rather than a summary of them. The determinations committee publishes its questions and rulings, and reading a contested one (Greece 2012 or the 2022 sovereign cases) conveys the character of the problem. On the auction mechanism, the academic work on its design and on whether the auction price is biased relative to eventual recovery is small, and the finding of Chernov, Gorbenko, and Makarov (2013) and Du and Zhu (2017) that biases exist and are related to the net open interest is worth knowing before relying on

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auction recoveries as data. On the Credit Suisse AT1 write-down, the Swiss regulator's own explanation and the subsequent statements from other supervisors are the documents to read, and the Federal Administrative Court's October 2025 judgment, now on appeal to the Federal Supreme Court, is the authoritative legal account so far. For the tranche and basket valuation of Sections 25.7 and 25.8, Hull and White (2004) and Andersen, Sidenius, and Basu (2003) are the sources of the four steps, and O'Kane and Turnbull (2003) is the practitioner treatment of the standard model. On convertibles, Tsiveriotis and Fernandes (1998) is the model behind most vendor systems, and the practitioner literature on convertible arbitrage sets out what happened in 2008.

PRACTICE QUESTIONS

25.1 Explain why a protection buyer need not own the reference obligation, giving one benefit and one problem this creates, and explain why gross CDS notional outstanding is not a measure of the risk transferred.

25.2 A five-year CDS has a par spread of 250 bp and trades on a 100 bp coupon with a risky annuity of 4.19. Compute the upfront payment, and explain why the risky annuity falls as the spread widens.

25.3 A desk bought five-year protection on the standard 100 bp coupon when the par spread was 150 bp, paying an upfront of 2.18% of notional. One year later the spread is 400 bp, and the risky annuity on the remaining four years at that level is 3.31. Compute the position's value and its gain since inception as percentages of notional, explain why marking against 150 bp or using the annuity at the original spread misstates them, and state in which direction the annuity error biases a hedge.

25.4 Explain why physical settlement failed, what the auction replaced it with, and why an auction recovery is not the same as the eventual realized recovery.

25.5 Explain why an index CDS trades below the arithmetic average of its constituents' spreads, what the index skew is, and why it widens when the index is used as a macro hedge.

25.6 Explain why restructuring creates a cheapest-to-deliver problem that bankruptcy does not, and why a risk system treating restructuring and no-restructuring contracts as substitutes has mismeasured the exposure.

25.7 State what the Credit Suisse AT1 write-down established about the meaning of subordination, and explain why CDS on Credit Suisse did not pay.

25.8 Explain why an equity tranche's expected loss falls as correlation rises while a senior tranche's rises, given that the portfolio's expected loss is unchanged, and use Example 25.5 to state the factor by which a 15–30% tranche's expected loss grows between $\rho = 0.10$ and $\rho = 0.50$.

25.9 Explain why implied correlation differing across tranches on the same portfolio is logically impossible, and what base correlation does about it.

25.10 A convertible has a conversion ratio of 25 and the stock trades at 30. Compute the conversion value and the conversion price on a face of 1000, and explain why the instrument behaves like a bond at low share prices and like equity at high ones.

25.11 State the four exposures of a delta-hedged convertible arbitrage position, and explain why two of them go wrong together and why the timing of their recognition differs.

25.12 Explain why a total return swap position does not appear in ownership disclosures, and who bears the risk if the receiver cannot meet a margin call.

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25.13 Distinguish the three causes of the CDS market's post-crisis shrinkage, and state which of them reduced risk rather than merely reducing gross notional.

25.14 Explain why a credit-linked note contains two credit exposures rather than one, and explain what the quanto spread between a dollar- and a euro-denominated CDS on a euro-area sovereign measures.

25.15 State the four residual risks a protection buyer faces even when right about the credit, and explain which of them is least measured and why it is the one most likely to produce the largest loss.

25.16 Five issuers each trade at 150 bp. Explain why the first-to-default basket costs the sum of the five spreads at $\rho = 0$ and falls as ρ rises while the fifth-to-default rises, and why the five baskets' protection legs add up to the five single-name legs (Example 25.6).

FURTHER QUESTIONS

25.17 Example 25.2 shows the risky annuity falling from 4.55 to 3.40 as the spread widens. Derive this dependence, show that it makes the upfront a concave function of the spread, and explain what this implies for a desk that hedges a portfolio of wide-spread issuers with a position in a tight-spread one.

25.18 Section 25.5 argues that the auction solved the squeeze and created a different problem. Set out the incentives of a participant with a large protection position, explain what the two-stage design does to constrain them, and propose a modification that would further reduce the incentive to influence the outcome, stating what it would cost.

25.19 Example 25.4 attributes the index-average gap to differences in risky annuity. Derive the exact relation between the index spread and the constituent spreads, show that the gap grows with dispersion, and determine the dispersion at which it becomes large enough to justify trading 125 issuers against the index.

25.20 Example 25.7 finds the restructuring clause worth 7% of the spread. Explain why this number is nearly useless as a risk measure, construct a measure that would capture the exposure correctly, and discuss why a bank hedging a loan book might rationally pay far more than 7% for the clause.

25.21 The Credit Suisse case study argues that where a payoff is defined by a legal event, the document is the instrument. Identify three other instruments in this book whose payoff depends on a legal or administrative determination rather than a market price, and for each state what the residual risk is and who is best placed to bear it.

25.22 Re-price Example 25.9's convertible with a 3% dividend yield on the stock and show that early conversion now has value; then add a holder's put at par at year 3 and state which way each feature moves the value and why.

25.23 Section 25.14 argues that convertible arbitrage was hedged in the model and failed in practice. Distinguish the model risk, the funding risk, and the crowding risk in that failure; explain which of the three the Greeks would have revealed; and relate your answer to Section 5.15 and to Chapter 22's account of 2018.

25.24 Using Example 25.10, show that the 80 bp band on the hybrid coupon does not depend on the index probability for any value between 0.10 and 0.90, and explain what that implies for how a desk should reserve against an unquoted correlation.

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